PROJECT SPECIFICATION

Trainable Face Recognition with Multi-Face Detection and Video Tracking

Four-week deep learning project | Python + PyTorch | Academic evaluation focus

1. Project overview

Develop a system that (1) detects faces in images and video, (2) tracks each face across consecutive frames using anonymous tracking IDs, and (3) recognizes only voluntarily enrolled individuals. The primary academic deliverable is a face recognition CNN that is actually trained, validated, tuned, and evaluated, not just an integration of pretrained software.

Central research question: How does a custom CNN trained from scratch compare with a fine-tuned pretrained ResNet18 for facial identity classification, and how reliably can the selected model support recognition in tracked video?

2. Scope and functional requirements

Priority	Requirement
Must have	Load images/videos; detect multiple faces; assign temporally stable track IDs; recognize enrolled identities or return Unknown.
Must have	Train two recognition baselines; save checkpoints; plot training/validation loss and accuracy; evaluate held-out test data.
Must have	Export annotated MP4/images and a CSV with frame, track ID, recognized identity, confidence, and bounding box.
Optional	Real-time webcam demo; appearance-assisted track reassociation after short occlusions.
Out of scope	Identifying strangers, operating on uncontrolled public surveillance footage, or training a large face detector from scratch.

3. System architecture

Input (image/video) -> Pretrained SCRFD face detector -> bounding boxes + landmarks -> alignment and quality checks -> ByteTrack (video track IDs) + trained CNN (identity probabilities/embeddings) -> similarity/threshold and temporal confirmation -> annotated output.

Detection and tracking are integration components. The recognition CNN is the trained model. A tracking ID is not a confirmed identity; Unknown is the default if there is insufficient evidence.

4. Datasets and data partitioning

Dataset	Role / notes
CelebA (202,599 images, 10,177 identities)	Primary candidate for supervised identity classification. Check identity annotation access and non-commercial research terms.
WIDER FACE (32,203 images)	Optional detector benchmark only; face bounding boxes, not person identities.
Controlled, consented recordings	5-10 enrolled people, 5-10 images each and 10-16 short videos; evaluate video tracking and unknown-person rejection.

Sampling: Choose approximately 100-300 adequately represented identities from CelebA, subject to available annotations. Split selected images by identity into stratified train (70%), validation (15%), and test (15%) partitions for known-identity classification. Remove near-duplicates across partitions. Also hold out a disjoint set of identities for unknown-identity rejection; this evaluates a different generalization claim. Keep recording sessions separate between enrollment and video testing.

5. Training and model selection

Experiment	Implementation
A - From scratch	Implement 3-5 convolutional blocks with batch normalization, ReLU, pooling and dropout; final softmax classifier.
B - Transfer learning	ImageNet-pretrained ResNet18; replace head; train new head, then unfreeze later layers and fine-tune.
Loss / optimization	Cross-entropy; AdamW; starting learning rates 1e-3 (custom CNN) and 1e-4 (fine-tuning), batch 32-64, 15-30 epochs.
Augmentations	Horizontal flip, mild brightness/contrast and small rotations; avoid unrealistic face transformations.
Validation	Record metrics per epoch; early stopping by validation loss; retain best checkpoint; do not tune using test set.

Open-set recognition: A classifier trained on fixed identities does not inherently generalize to unseen enrolled people. For the month-one system, enroll identities already represented in the trained class set and reject unknowns using calibrated confidence plus a validation-based threshold. If time permits, assess penultimate-layer embeddings on entirely unseen identities, clearly reporting this as a separate experiment.

6. Evaluation protocol and required figures

• Classification: test accuracy, macro precision/recall/F1, confusion matrix and per-class errors.

• Training: train/validation loss curves, accuracy curves, learning-rate sensitivity, augmentation ablation, overfitting discussion.

• Unknown rejection: false accept rate, false reject rate, and TAR at a specified FAR, measured with separate unknown identities.

• Video tracking: ID switches, IDF1 where annotated ground truth exists, throughput in frames/second; test crossing and short occlusions.

• Integrated: track-level identity accuracy and number of wrong confirmed labels on annotated, consented recordings.

Controlled comparison: fixed dataset partitions and comparable training budgets for A and B. Optional ablation: per-frame predictions vs temporal voting. Clearly report mean/variance if repeating runs with multiple random seeds.

7. Four-week schedule

Week	Tasks	Deliverable
1	Acquire labels; audit examples; make splits; train scratch CNN.	Baseline notebook and first curves
2	Fine-tune ResNet18; tune learning rate, dropout, augmentation.	Best checkpoint, experiment log and comparison
3	Hold-out evaluation; integrate SCRFD, ByteTrack and identity association.	Annotated video and error analysis
4	Create Gradio UI; benchmark full pipeline; finalize report and demo.	Working app, results, README and report

8. Deliverables and acceptance criteria

- PyTorch training scripts/notebooks with reproducible seeds, dataset split logic and training configuration.
- Two trained checkpoints and experiment logs showing meaningful training/validation analysis.
- Metrics report with confusion matrix, learning curves, recognition rejection analysis and tracking examples.
- Application supporting video/image upload and export; GitHub repository without sensitive biometric data.
- Final write-up explaining architecture, losses, optimizer, regularization, model selection and limitations.

9. Risks, ethics and fallback

Risks: limited images per identity, dataset access, face blur/occlusion, class imbalance, training time and accidental train/test leakage. Fallback: prioritize a fully evaluated face classifier and image inference first; tracking is the second milestone, webcam inference optional. Enroll only consenting participants, permit deletion of photos/embeddings, avoid public identification, and verify license terms of all datasets and pretrained detector weights. In EU settings, facial identification involves sensitive biometric data and requires appropriate legal safeguards.

10. Technology and source links

Stack: Python, PyTorch, torchvision, OpenCV, InsightFace/SCRFD, ByteTrack, scikit-learn, Matplotlib, Gradio; optional Colab/Kaggle GPU.

CelebA: https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html
WIDER FACE: https://mmlab.ie.cuhk.edu.hk/projects/WIDERFace/
InsightFace: https://github.com/deepinsight/insightface
ByteTrack: https://github.com/ifzhang/ByteTrack